

Arielle C. Frommer

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EDUCATION

Yale University

Ph.D. in Astronomy (expected 2031)

New Haven, CT

September 2026–present

Harvard University

A.B. in Astrophysics & Physics

Cum Laude with Highest Honors in Field — GPA: 3.9/4.0

Cambridge, MA

September 2021–May 2025

AWARDS AND FELLOWSHIPS

Yale Gruber Science Fellowship, Yale University

2026

Graduate Research Fellowship Program, National Science Foundation

2026

Alex G. Booth Fellowship, Harvard University

2025

Chambliss Astronomy Achievement Student Award, 246th AAS meeting

2025

Leo Goldberg Prize for best junior thesis, Harvard Astronomy Department

2024

Summer Program for Undergraduates in Data Science, Harvard University

2022

PUBLICATIONS

1. **Frommer, A. C.**, J. Eastman, D. Ciardi, et al. A Paradigm Shift in Exoplanet False Positive Identification with HIP-44302. 2025, submitted to [AJ](#).
2. Heidari, N., Hébrard, G., Martioli, E., et al. **incl. Frommer, A. C.** Characterization of seven transiting systems, including four warm Jupiters from SOPHIE and TESS. 2025, [A&A](#), 694, A36.
3. Law, C. J., Zhang, Q., **Frommer, A. C.**, et al. A Wideband Chemical Survey of Massive Star-forming Regions at Subarcsecond Resolution with the Submillimeter Array. 2024, [ApJS](#), 276, 54.

RESEARCH

Leiden Observatory

Visiting Researcher

Project: Measuring $^{12}\text{CO}/^{13}\text{CO}$ in WASP-39b's JWST Transmission Spectrum

Advisor: Prof. Ignas Snellen

Leiden, the Netherlands

September 2025–April 2026

Center for Astrophysics | Harvard & Smithsonian

Senior Thesis

Project: Using Machine Learning to Improve TESS's Systematic Error

Advisors: Dr. Jason Eastman and Prof. Andrew Vanderburg

Cambridge, MA

September 2024–May 2025

Junior Thesis

Project: Modeling the Triple-Star System TOI-568 Using EXOFAST

Advisor: Dr. Jason Eastman

September 2023–May 2024

Student Researcher

Project: Probing Nitrile Chemistry in Massive Star Formation

Advisors: Dr. Qizhou Zhang and Dr. Charles Law

June 2022–May 2023

University of Washington Institute for Nuclear Theory

NSF REU

Project: Simulating Neutrino Many-Body Systems

Advisors: Dr. Vincenzo Cirigliano and Dr. Yukari Yamauchi

Seattle, WA

June 2024–August 2024

PRESENTATIONS

Talks:

Measuring $^{12}\text{CO}/^{13}\text{CO}$ in WASP-39b's JWST Transmission Spectrum March 2026
Leiden Exoplanets Meeting, Leiden, the Netherlands

Modeling the Triple-star System TOI-568 Using EXOFAST January 2024
243rd AAS meeting, New Orleans, LA

A Survey of Nitrile Chemistry in Massive Star-Forming Regions January 2023
241st AAS meeting, Seattle, WA

Posters:

Measuring $^{12}\text{CO}/^{13}\text{CO}$ in WASP-39b's JWST Transmission Spectrum July 2026
ERES XI, Ann Arbor, Michigan

Using Machine Learning to Improve TESS's Systematic Error June 2025
246th AAS meeting, Anchorage, AK

Using Machine Learning to Improve TESS's Systematic Error January 2025
245th AAS meeting, National Harbor, MD

WORK EXPERIENCE

Sky & Telescope Cambridge, MA
Freelance journalist April 2024–present
Writes news and feature pieces covering astronomy discoveries for online blog and print magazine

Harvard Art Museums Cambridge, MA
Ho Family Student Guide September 2022–May 2025
Developed and gave original tours on art & astronomy and art & poetry for the general public

ACTIVITIES & OUTREACH

Mystic Seaport Treworgy Planetarium Mystic, CT
Planetarium Operator June 2026–present
Leads live planetarium shows about tonight's night sky, weaving together science, mythology, and celestial navigation

Astronomy on Tap Leiden Leiden, the Netherlands
Committee Member October 2025–present
Helps organize monthly public talks by astronomers at local pubs to promote astronomy outreach

Harvard Undergraduate Astronomy Society Cambridge, MA
Co-President (2024–2025), Co-Social Chair (2023–2024) May 2023–May 2025
Planned weekly events, informative panels, and mentorship initiatives to foster community for undergraduate astronomy students

Harvard Crimson Arts Board Cambridge, MA
Editor-at-Large (2024), Culture Editor (2023) January 2023–December 2024
Wrote and edited reviews and articles for the Arts section of The Harvard Crimson student newspaper

Harvard Advising Programs Office

Peer Advising Fellow

*Mentored 8-10 freshman advisees and organized weekly study breaks within their dorm to build community***Cambridge, MA**

September 2022–May 2024

Harvard First-Year Outdoor Program and Harvard Outing Club

Pre-Orientation Leader

*Led backpacking trips for freshmen prior to orientation and day trips during the year to foster community and appreciation for the outdoors***Cambridge, MA**

January 2023–May 2025

PBHA Franklin Afterschool Enrichment Program

Volunteer

*Tutored elementary school students in Dorchester and organized engaging activities and field trips***Cambridge, MA**

September 2021–May 2022

INTERESTS

Research Interests: exoplanets, planet formation, habitability, astrochemistry, machine learning**Coding languages:** Python, Unix/Linux, IDL, Git/Github, Excel, DS9, IRAF, CASA**Hobbies:** science writing, creative writing, hiking, dance, reading, piano, crocheting